CO7063

Linux Command Line & Kali Linux in VirtualBox

Combined workshop summary and step-by-step practical Notes

What this guide combines

Workshop A: “Setting Up a Kali Linux VM in VirtualBox”
Workshop B: “Linux Lab 1 - Command Line Boot Camp”.

The goal is to make the setup process and Linux commands easier to understand and remember..

Prepared from the two workshop Slides:

1. Setting Up a Kali Linux VM in VirtualBox
2. Linux Lab 1 (Command Line Boot Camp)

Notes Overview:

Section	What you will learn / do
1. Learning objectives	p. 2 - Understand the expected skills and how the two handouts fit together.
2. Before you begin	p. 3 - Check hardware, disk space, internet and virtualisation requirements.
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4. Linux terminal basics	pp. 6-7 - Understand the prompt, paths, home folder, hidden files and manual pages.
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6. Evidence and report checklist	p. 14 - Know which findings and screenshots to save for the lab document.
7. Troubleshooting and safety	p. 15 - Resolve common setup problems and avoid accidental deletion.
8. Revision	pp. 12-13 and 16-18 - Use the command reference, glossary, self-test and final recap.

Important: the two handouts use different Linux distributions

Do not confuse Kali and Fedora

The VirtualBox setup handout explains how to set up a Kali Linux virtual machine. The Linux Lab 1 handout explicitly says to use a Fedora virtual machine. The commands in Lab 1 are common Linux commands and are usually similar on both systems, but follow your tutor's requirement about which VM to use. Do not assume Kali replaces Fedora for submission unless your tutor says so.

Learning objectives

- Prepare a suitable computer and install VirtualBox.
- Download, extract, add and start a pre-built Kali virtual machine.
- Log in, test network connectivity and understand optional first-run maintenance steps.
- Open a terminal and identify the current working directory.
- Navigate absolute and relative paths, including the home directory and parent directory.
- List normal and hidden directory entries and read basic command help.
- Create directories and files, edit text, and inspect file listings.
- Copy and move files using a destination path and use a simple wildcard.
- Delete a file safely and explain why removal commands require care.
- Produce a concise lab record with findings and requested screenshots.

PART 1

Before you begin

These requirements and links come from the Kali setup handout. Check your own computer before downloading several gigabytes of files.

Requirement	Handout guidance	Why it matters
Computer	Windows, macOS or Linux desktop/laptop	VirtualBox runs on the host computer.
Memory (RAM)	8 GB recommended; 4 GB minimum, but slow	The host operating system and virtual machine share physical memory.
Free disk space	At least 40 GB	The downloaded archive, extracted VM and running VM all need storage.
Internet	Stable connection	The VM download is several GB, and the sanity check needs internet access.
Hardware virtualisation	Enabled in BIOS/UEFI; usually already enabled	VirtualBox may refuse to start the VM if Intel VT-x / AMD-V / SVM is disabled.
Extraction utility	For example, 7-Zip on Windows	The downloaded Kali VM is supplied in a .7z archive.

Useful words

Term	Plain-English meaning
Host computer	Your real laptop or desktop.
Virtual machine (VM)	A computer running as software inside your real computer.
VirtualBox	The application that creates and runs virtual machines.
Guest operating system	The system running inside the VM - in this workshop, Kali Linux or (for Lab 1) Fedora.
Terminal	A text-based window where you type Linux commands.
Archive (.7z)	A compressed package that must be extracted before using the VM files.

Official links supplied in the handout

- VirtualBox downloads: https://www.virtualbox.org/wiki/Downloads
- Kali pre-built virtual machines: https://www.kali.org/get-kali/#kali-virtual-machines
- Kali: import a pre-made VirtualBox image: https://www.kali.org/docs/virtualization/import-premade-virtualbox/
- 7-Zip for Windows: https://www.7-zip.org

Version note: the provided handout gives kali-linux-2026.2-virtualbox-amd64.7z as an example archive name. File names and available versions can change; use the actual filename you downloaded.

PART 2

Set up Kali Linux in VirtualBox

Follow these steps once to prepare the environment for practical workshops.

Step 1 - Install VirtualBox

1. Open the official VirtualBox download page listed above.
2. Under “VirtualBox Platform Packages”, choose the package for your host operating system. On Apple Silicon Macs, choose the Apple Silicon package rather than the Intel package.
3. Run the downloaded installer and follow the setup wizard, accepting the default options unless your institution instructs otherwise.
4. On Windows, approve the prompts for installing network adapters if they appear.
5. Launch VirtualBox once. Confirm that the main window opens without an error.

Extension Pack

The handout says the Extension Pack is optional for this workshop. It adds features such as USB 3.0 support; skip it unless you need it and know why.

Step 2 - Download the pre-built Kali VM

1. Open the official Kali virtual-machines page.
2. Select the VirtualBox option/tab and download the 64-bit VirtualBox image. The handout describes it as a large .7z archive of roughly 3-4 GB.
3. Wait for the download to finish before extracting it.

Step 3 - Extract the downloaded archive

Host operating system	Method described in the handout
Windows	Install 7-Zip if needed. Right-click the .7z file and choose 7-Zip > Extract Here (or extract to a folder).
macOS	Use The Unarchiver or Keka, then extract the archive.
Linux	Open a terminal in the download folder and use the example command below, adjusting the name if yours differs.

```
7z x kali-linux-2026.2-virtualbox-amd64.7z
```

Example filename from the handout only. Replace it with the actual downloaded filename.

Extraction can take several minutes and temporarily requires additional disk space - the handout advises allowing roughly double the download size.

Step 4 - Add the extracted VM to VirtualBox

1. Open VirtualBox.
2. Click Add in the main toolbar (sometimes shown as a folder with a plus sign).
3. Browse to the extracted Kali folder and select the .vbox file inside it. This file stores the VM configuration; this step adds a pre-built VM rather than installing from an ISO image.
4. Confirm that Kali appears in the VM list on the left.
5. Select the VM and review its settings. The handout says the defaults are fine for the workshop.
6. If the host has 8 GB RAM or more, you can consider setting the VM base memory to around 4096 MB under Settings > System > Motherboard. Do not assign more memory than the host can spare.

PART 2 CONTINUED

First boot, login and sanity check

Do the first boot before the Linux lab so that you know the VM starts and can access the network.

Step 5 - Start Kali and log in

1. Select the Kali VM in VirtualBox and click Start.
2. Wait for the login screen. The first boot may take a minute or two.
3. Use the default credentials stated in the handout: username kali and password kali. If your image shows different credentials, follow the information supplied with that image or by your tutor.
4. Optional but recommended: after logging in, open a terminal and change the default password.

```
passwd
```

Starts the password-change process for your current Linux account. Follow the prompts and choose a strong password.

Step 6 - Take a clean snapshot (optional)

In VirtualBox, use Machine > Take Snapshot and name it “Clean Install”. A snapshot is a restore point for the VM state, useful if a workshop change breaks the environment. It is not a substitute for a separate backup of important files.

Step 7 - Test network connectivity

```
ping -c 4 kali.org
```

Sends four test packets to check whether the VM can reach kali.org. A successful result should show replies and packet statistics. A failed ping can also reflect DNS, network or host filtering issues.

Step 8 - Optional package update

```
sudo apt update && sudo apt upgrade -y
```

Refreshes package lists and then upgrades installed packages. This may take a while. The handout says it is optional and can be done before the first workshop rather than during it.

What the command means

sudo runs a command with elevated privileges; apt update refreshes information about available packages; && runs the next command only if the first succeeds; apt upgrade -y applies available package upgrades and answers yes to prompts. Run this only in your own workshop VM and allow enough time and disk space.

PART 3

Linux terminal basics

The Linux Lab 1 handout teaches commands for navigating folders and managing files. Commands are typed at the prompt and executed by pressing Enter.

1. Understand the prompt and current location

The terminal prompt often displays your login name and the computer name. You are always “inside” a working directory when running commands, unless a command is designed to work differently.

```
pwd
```

Print working directory: shows the full path of the directory you are currently in. Record this at the start of Lab 1.

2. Paths: /, ~, . and ..

Symbol / command	Meaning	Example
/	The root of the Linux filesystem - the highest-level directory.	cd /
~	Your home directory.	cd ~
.	The current directory.	ls .
..	The parent directory, one level above the current one.	cd ..
Absolute path	A full path starting from /.	/home/kali/Documents
Relative path	A path interpreted from your current working directory.	Documents/Labs

3. The main navigation and listing commands

Command	What it does	Remember
pwd	Prints the full current directory path.	Use it when unsure where you are.
ls	Lists visible items in the current directory.	It can also list a named directory.
ls /	Lists the contents of the root directory.	The slash is the root directory.
ls ..	Lists the contents of the parent directory.	Two dots mean “one level up”.
ls -a	Includes hidden entries, including names that begin with a dot.	-a means “all”.
ls -l	Shows a long listing with details such as permissions, owner, size and modification time.	Lowercase L, not number 1.
ls -F	Adds a symbol to indicate item types, such as / after directory names.	Exact symbols may vary by environment.
cd path	Changes to the named directory.	The path must exist.
cd ..	Moves up to the parent directory.	Can use repeatedly to move up.
cd ~ or cd	Takes you to your home directory.	When at home, the prompt may show ~.
cd /	Moves to the root directory.	This is not the same as home.

4. Hidden files and the meaning of dot entries

- A single dot (.) refers to the current directory.
- Two dots (..) refer to the parent directory.
- A filename that begins with a dot (for example, .config) is treated as hidden by common Linux listing tools. It may not appear in plain ls output.
- Use ls -a to display hidden entries as well as ordinary visible entries.

5. Use the manual to get help

```
man ls
```

Opens the manual page for ls. Use the arrow keys or Space to move through it; press q to quit.

The lab asks you to look up what -a and -l do. Write a one-sentence explanation of each in your own words.

Quick practice - check where you are

Run these commands one at a time. After each pwd, write down the path shown by your own VM.

```
pwd
```

Record your starting directory.

```
ls -a
```

Find the . and .. entries and any other dot-prefixed names.

```
cd /
```

Move to the root directory.

```
pwd
```

Confirm the path now shown.

```
cd ~
```

Return to your home directory.

```
pwd
```

Confirm the home-directory path. Compare it with the root path.

PART 4

Linux Lab 1 - step-by-step practical

The sequence below follows the intent of the supplied lab handout, with small clarifications where the source wording is inconsistent. Run commands one at a time and check the output before continuing.

Task 1 - Find your working directory

```
pwd
```

Look at the full path printed in the terminal.

Record: What is the path to your home directory? In many Linux installations it resembles /home/username, but use the path shown on your own VM.

Task 2 - List directory contents

```
ls
```

Lists visible entries in your current directory. Take the requested screenshot and save it in your Linux Lab document.

```
ls /
```

Lists the root-level directories.

```
ls ..
```

Lists the parent directory of your current location.

```
ls -a
```

Shows hidden entries, including . and .. and other dot-prefixed names.

In your report, explain the meaning of . and .., and why some file names begin with a dot. The handout asks you to record your findings.

Task 3 - Change directories

```
cd ..
```

Move up one directory level.

```
cd /bin
```

Move to /bin, a directory that commonly contains essential executable programs.

Take a screenshot showing the command lines, as requested in the handout.

Task 4 - Compare ways to return home

Command	Destination	What to notice
cd /	Root directory.	The prompt usually no longer shows the home-directory tilde.
cd ~	Your own home directory.	The tilde explicitly represents home.
cd	Your own home directory.	With no path given, cd normally returns you home.

Run each command separately and use pwd after each one to record the difference.

Task 5 - Read the command manual

```
man ls
```

Read the descriptions of -a and -l. Press q to exit the manual.

Report in your own words: What additional items does -a show? What extra detail does -l show?

PART 4 CONTINUED

Create folders and a text file

The handout alternates between “Material” and “Materials”, and between “ForensicsLab” and “ForensicsLabs”. This guide consistently uses “Materials” and “ForensicsLabs”.

Task 6 - Create the directory structure

Target layout: Documents contains Labs and Lectures; Labs contains ForensicsLabs, Questions and Answers; Lectures contains Materials. A Forensics subdirectory goes inside ForensicsLabs.

```
cd ~
```

Start from your home directory.

```
mkdir -p Documents/Labs/ForensicsLabs Documents/Labs/Questions Documents/Labs/Answers Documents/Lectures/Materials
```

Creates the listed directories, including any missing parent folders. -p allows parent folders to be created as needed.

```
cd ~/Documents/Labs/ForensicsLabs
```

Move into the working directory used by the later tasks.

```
ls
```

Check the contents of the working directory. It may be empty at this point.

```
mkdir Forensics
```

Create the Forensics subdirectory inside ForensicsLabs.

The handout also demonstrates `mkdir -p Lectures/Materials` as a nested-directory example; the combined command above has already created this path. To practise that exact relative-path example separately, first run `cd ~/Documents`.

Take a screenshot of your command lines and the created directory structure, as the handout requests.

Task 7 - Create and edit music.txt

```
cd ~/Documents/Labs/ForensicsLabs
```

Make sure you are in the ForensicsLabs directory.

```
touch music.txt
```

Creates an empty file called music.txt. If it already exists, touch updates its timestamp rather than emptying it.

```
nano music.txt
```

Opens the Nano text editor. Type two or three favourite bands or album names.

To save and exit Nano: press Ctrl+X, press Y to confirm saving, then press Enter to accept the filename.

```
ls
```

Check that music.txt appears in the current directory.

Task 8 - Make ls more useful

```
ls -F
```

Displays entries with type-indicating suffixes. A slash often identifies a directory.

```
ls -l
```

Shows a long listing with more information, including ownership and modification time. Take the screenshot requested in the handout.

Record what you notice about files versus directories. Colours vary between terminals, so do not rely on colour alone.

PART 4 CONTINUED

Copy, move and remove files

These tasks practise the difference between making a second copy and relocating an existing file.

Task 9 - Copy a file

Before starting, confirm you are in ForensicsLabs and that music.txt exists.

```
pwd
```

Confirm your working directory.

```
ls
```

Check that music.txt is present.

```
cp music.txt copymusic.txt
```

Creates a copy in the same directory with a new name: copymusic.txt.

```
cp music.txt Forensics/
```

Copies music.txt into the Forensics directory, keeping the filename music.txt.

```
cp music.txt Forensics/my_music.txt
```

Copies music.txt into Forensics and names the copy my_music.txt.

```
ls -F Forensics/
```

Check which copies are inside Forensics/. Take the screenshot requested for this task.

Task 10 - Move a file and practise a wildcard

```
mv copymusic.txt Forensics/
```

Moves copymusic.txt from ForensicsLabs into the Forensics directory. The source file is no longer in the original location.

```
mv *.txt Forensics/
```

Moves any remaining .txt files in the current directory into Forensics/. The asterisk * is a wildcard meaning “match any sequence of characters”.

```
ls -l Forensics/
```

Inspect the destination directory and confirm which files arrived.

Important difference

cp copies: the original remains. mv moves/renames: after a successful move, the item is no longer at its old path. A wildcard such as *.txt can match more than one file, so always run ls first and understand which files will be affected.

Task 11 - Remove a file or directory

The handout says to remove copymusic.txt from inside Forensics. You must be in that directory or give the full path. The following steps make the location explicit:

```
cd ~/Documents/Labs/ForensicsLabs/Forensics
```

Move into the Forensics directory.

```
pwd
```

Verify the current location before deleting anything.

```
ls
```

Confirm that copymusic.txt is present. If it is not there, do not run the removal command blindly; check where it went.

```
rm copymusic.txt
```

Deletes that file. There is typically no recycle bin or “undo” command in the terminal.

```
ls
```

Check that copymusic.txt has been removed.

```
rmdir EMPTY_DIRECTORY_NAME
```

Removes a directory only if it is empty. Replace EMPTY_DIRECTORY_NAME with the actual directory name; this is a syntax example, not a command to copy literally.

Deletion safety rule

Before rm or mv, check pwd and ls. Avoid practising with important personal files. Do not add wildcards to rm unless you have checked exactly what they match. The source handout specifically warns that recovery after rm can be difficult or impossible.

PART 5

Command reference at a glance

Keep this page nearby while completing the lab. The commands are shown as used in the supplied workshop handouts.

Command	Meaning in plain English	Example / note
pwd	Show where you are.	pwd
ls	List visible entries.	ls / lists the root directory.
ls -a	List all entries, including hidden dot-prefixed names.	ls -a
ls -l	List details in long format.	ls -l
ls -F	Mark file types with suffixes where supported.	ls -F
cd NAME	Change to a directory.	cd /bin
cd ..	Move to parent directory.	cd ..
cd ~ / cd	Return to your home directory.	cd ~
man COMMAND	Open the manual page.	man ls, then q to exit.
mkdir NAME	Create a directory.	mkdir Forensics
mkdir -p A/B	Create missing parent directories too.	mkdir -p Lectures/Materials
touch FILE	Create an empty file if it does not exist.	touch music.txt
nano FILE	Open a text file in Nano, creating it if needed.	nano music.txt
cp SOURCE DEST	Copy a file to a new file or directory.	cp music.txt copymusic.txt
mv SOURCE DEST	Move or rename a file.	mv copymusic.txt Forensics/
rm FILE	Delete a file.	rm copymusic.txt
rmdir DIR	Delete an empty directory.	rmdir empty_folder
*	Wildcard matching any string of characters.	*.txt matches names ending in .txt.

Mini mental model: how to read a command

Many commands follow this shape: command [options] [source] [destination]. For example, cp music.txt Forensics/my_music.txt means “run cp; use music.txt as the source; create a destination copy named my_music.txt inside Forensics.” Not every command uses all these parts.

Common options and symbols

Option / symbol	Plain-English meaning
-a	Show all entries, including hidden ones, for ls.
-l	Use the long, detailed listing format for ls.
-F	Add a suffix that helps indicate item type for ls.
-p	With mkdir, create parent directories as needed.
~	Home directory.
/	Root directory, or a separator between directory names.
..	Parent directory.
*	Wildcard matching any sequence of characters.
Ctrl+X, Y, Enter	Save and exit Nano in the workflow described by the handout.
-c	With ping, send the specified number of packets.
sudo	Run a command with elevated privileges.
&&	Run the command on the right only if the command on the left succeeds.
-y	For apt upgrade in the handout, automatically answer yes to confirmation prompts.

PART 6

Lab report and screenshot checklist

The Linux Lab handout repeatedly asks you to capture terminal screenshots and write down findings. Use this checklist to avoid leaving evidence until the end.

Evidence to include

Lab task	Evidence / finding to record
Task 1	Record the full path shown by pwd and identify your home directory.
Task 2	Screenshot of ls; written explanation of . , .. and dot-prefixed hidden names; note what ls -a shows.
Task 3	Screenshot showing cd .. and cd /bin.
Task 4	Record the differences among cd /, cd ~ and cd. Use pwd after each command.
Task 5	Explain ls -a and ls -l using information from man ls.
Task 6	Screenshot of the directory-creation commands and check the resulting folder structure.
Task 7	Show that music.txt exists; explain that you edited it in Nano and how you saved it.
Task 8	Screenshot of ls -l and describe the extra information displayed.
Task 9	Screenshot/listing confirming copymusic.txt, Forensics/music.txt and Forensics/my_music.txt were created.
Task 10	Record how mv changed the location of files and explain what *.txt matched.
Task 11	Record the removal of copymusic.txt and explain why rm must be used carefully.

Simple report structure

1. Title: CO7063 Linux Lab 1 - Command Line Boot Camp.
2. Aim: one or two sentences describing the purpose of the lab.
3. Environment: Linux distribution and whether it ran in a VM; include the VM name if useful.
4. Method and evidence: list the task number, command used, screenshot, and short explanation of the result.
5. Reflection: describe three commands you understand better now and one issue you had to solve.
6. Conclusion: state whether you completed the objectives and what you learned.

Good screenshot practice

Capture the relevant terminal output with enough context to show the command and result. Do not crop away the command prompt if it helps prove which directory you were in. Avoid including passwords or unrelated personal information.

PART 7

Troubleshooting

Use the symptom to narrow down the likely cause. The first four issues below come directly from the Kali setup handout; the Linux command notes are practical consequences of the lab steps.

Problem	Possible cause	What to try
“VT-x is disabled” or similar	CPU virtualisation is disabled in BIOS/UEFI.	Restart and enter BIOS/UEFI (often F2, F10, F12 or Del). Look for Intel VT-x, AMD-V or SVM Mode and enable it, following your device maker’s instructions.
VirtualBox reports a Hyper-V conflict on Windows	Hyper-V or Core Isolation / Memory Integrity may conflict with VirtualBox.	The handout suggests trying Memory Integrity under Windows Security > Device Security first, restarting, and retrying. If issues persist, its next step is to review Hyper-V and related Windows features. These changes affect host security; follow institutional IT guidance before changing them.
VM is very slow	Not enough RAM/CPU assigned or too many host applications are running.	Close resource-heavy applications; consider increasing VM RAM and CPU cores under Settings > System, within the host computer’s limits.
No space during extraction/import	Archive extraction requires additional temporary disk space.	Free disk space or extract to a secondary/external drive; allow extra space beyond the archive size.
Command says “No such file or directory”	A path is mistyped or you are in a different directory.	Run pwd and ls; check spelling and capitalisation, then use the right relative or absolute path.
“File exists” after mkdir	You already created that directory.	Run ls to confirm it exists; continue rather than creating a differently named duplicate.
Nano changes did not save	Save steps were skipped.	Open nano music.txt, press Ctrl+X, press Y, and press Enter when prompted for the filename.
cp or mv cannot find music.txt	The file is not in the current directory or has a different name.	Run pwd and ls, then move to ForensicsLabs or specify the correct path.
ping fails	The VM may not have internet/DNS connectivity, or ping may be filtered.	Check VirtualBox networking and host internet. Record the actual message; do not assume a failed ping alone proves every network function is down.
rm says the file is missing	You may be in the wrong directory or the file has already been moved/deleted.	Run pwd and ls and locate the file before trying another deletion.

Before asking for help, capture these details

- Host operating system (Windows/macOS/Linux) and device type.
- VirtualBox error text exactly as shown.
- Which step you were following and the command typed.
- Output of pwd and ls, if the issue is with a Linux file path.
- Whether the VM has been started before and whether a snapshot exists.

PART 8

Glossary and self-test

Use the glossary for quick revision, then answer the questions without looking at the command sheet.

Glossary

Term	Easy definition
BIOS / UEFI	Firmware settings that control low-level computer hardware options, including virtualisation.
Command line	A text interface for entering instructions.
Current working directory	The directory the terminal is currently operating from.
Directory	A folder used to organise files and other directories.
File path	The sequence of directory names locating a file or directory.
Hidden file	A file whose name starts with a dot; ordinary ls may not show it.
Home directory	A user's personal starting directory, represented by ~.
Parent directory	The directory one level above the current one, represented by .. .
Root directory	The top level of the Linux filesystem, represented by /.
Snapshot	A saved point in the VM's state that can help revert workshop changes.
Wildcard	A pattern symbol, such as *, that can match multiple filenames.
Package list	Information about software packages available from configured repositories.

Self-test questions

1. Which command prints your full current directory path?
Answer: pwd.

2. What is the difference between cd / and cd ~?
Answer: cd / goes to the root directory; cd ~ goes to your home directory.

3. Why might ls not show a file beginning with a dot?
Answer: Dot-prefixed names are hidden by default in ordinary ls output; use ls -a.

4. How do you read the manual for ls, and how do you exit it?
Answer: Run man ls; press q to quit.

5. What does mkdir -p help with?
Answer: Creating nested directories while also creating any missing parent directories.

6. What is the difference between touch music.txt and nano music.txt?
Answer: touch creates an empty file if missing (or updates its timestamp); nano opens a text editor and can create/edit the file.

7. What is the difference between cp and mv?
Answer: cp makes a copy; mv moves or renames the original.

8. What does *.txt mean in the lab?
Answer: It matches names ending in .txt in the current directory.

9. Why should you check pwd and ls before rm?
Answer: To confirm your location and the exact files before an operation that may be difficult to undo.

10. What does ls -l add compared with plain ls?

Answer: A detailed listing such as permissions, ownership, file size and modification time.

11. Which VM does the Linux Lab 1 handout explicitly request?

Answer: Fedora. The separate setup handout explains setting up Kali; ask the tutor if unsure which VM to submit work from.

12. What network check is used in the Kali handout?

Answer: ping -c 4 kali.org.

13. What are the default credentials written in the Kali setup handout?

Answer: Username kali and password kali, for the image described in that handout; change the default password after login.

14. What should you do if mv or cp says a file is missing?

Answer: Check pwd, list the files with ls and verify the path/spelling.

15. Can rmdir remove a directory containing files?

Answer: No. The handout says rmdir only removes an empty directory.

FINAL RECAP

The practical workflow in one view

A short sequence to remember before a workshop or revision session.

Stage	Do this	Success looks like
1. Prepare	Check RAM, disk space, internet and hardware virtualisation.	The host computer is ready to run a VM.
2. Install	Install VirtualBox from the official page.	VirtualBox opens.
3. Download	Download and extract the pre-built Kali VirtualBox archive.	The extracted folder contains the VM configuration file.
4. Add and boot	Add the .vbox file, start Kali and log in.	Kali reaches the desktop.
5. Verify	Run ping -c 4 kali.org; optionally update packages.	You have a record of network-test output or failure.
6. Navigate	Use pwd, ls and cd.	You know where you are and how to move.
7. Build	Create the directory structure; use touch and nano.	Folders and music.txt exist.
8. Manage files	Use ls -a/-l/-F, cp and mv.	You can explain what changed and where each file is.
9. Clean up	Check pwd and ls before rm; use rmdir only for empty directories.	The intended file is removed without deleting unrelated work.
10. Report	Save screenshots and explanations for each requested task.	Your lab record shows both commands and understanding.

Three rules to take into every Linux lab

1. Know where you are: use pwd when uncertain.
2. Check what is there: use ls before copying, moving or deleting.
3. Understand the destination: cp duplicates; mv relocates or renames; rm deletes.

Source notes and limits

This guide is based on the two supplied documents: “Setting Up a Kali Linux VM in VirtualBox” (3 pages) and “Linux Lab 1 - Command Line Boot Camp” (6 pages). The handouts do not provide screenshots of your own VM, the actual output of commands on your machine, or tutor-specific submission formatting. Complete those items during your own practical. Where naming differs within the Linux Lab handout, this guide flags the inconsistency and uses one consistent path for the example workflow. The sentence explaining `cp music.txt Forensics/my_music.txt` also contains a typo in the displayed filename; the command itself specifies `my_music.txt`.

The setup handout describes one Kali VM image and its default login. Image names, credentials and software interfaces can vary by release or institutional build; follow the image-specific instructions if they differ.